MILITARY SPECIFICATION

CLOTH, DUCK, AROMATIC POLYAMIDE, NON-MELTING

1. SCOPE

1.1 Scope. This specification covers requirements for non-melting, aromatic, nylon cloth woven of plied yarns for use in the manufacture of seat cushion covers, armor vest covers, and for coating. One type of cloth is described in this specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-1133	Packaging and Packing of Synthetic Fiber Fabrics

Military

MIL-C-7020	Cloth, Parachute, Nylon

STANDARDS

Federal

Fed Std No. 4	Glossary of Fabric Imperfections
Fed Std No. 191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580).

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.2 First article. This specification contains provisions for first article inspection and approval (see 4.2, 6.2, and 6.6).

3.3 Materials

3.3.1 Fiber. The fiber shall be a 1.5 to 2 denier by 1.5 to 2 inch, non-melting, aromatic, polyamide fiber of sufficient strength to produce a spun and plied yarn with properties so that the fabric prepared from this yarn shall meet the physical properties as outlined in table I.

3.3.2 Yarn. The yarn shall be a two-ply yarn, smooth and free from spinning defects so that when converted into fabric shall yield fabric properties as defined in table I.

3.4 Construction and physical properties. The construction and physical properties shall be as specified in table I.

3.5 Color. The color shall match Sage Green 1565 or the standard sample and shall be acquired through the use of solution dyed fibers in this fabric.

3.6 Finish. The cloth shall be scoured and heat set and supplied with an antistatic finish.

3.7 Blocking. Unless otherwise specified, there shall be no sticking of cloth to cloth.

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3.8 Dimensions

3.8.1 Width. The width shall be as specified by the procuring activity.

Table I. Physical properties

Characteristics	Requirements
Weave	Plain, 1 up and 1 down
Weight (oz/yd^2), minimum	7.0
Thread count (W X F), minimum	47 X 47
Breaking strength (lbs: W X F), minimum	150 X 150
Tearing strength (lbs: W X F), minimum	20 X 20

3.8.2 Length of cut. Unless otherwise specified, the cloth shall be delivered in rolls. Pieces shall be 80 yards or more in length. Pieces less than 80 yards will be acceptable up to 10 percent of the contract quantity, provided no piece is less than 25 yards in length. Short pieces shall be rolled separately. Not more than two pieces less than 80 yards in length shall be included in any one roll unless otherwise specified in the contract. The pieces shall be sewn together with a butted type seam, using approximately 10/3 cord cotton thread, and have approximately nine stitches per inch. The stitching shall be placed not closer than one-half of an inch from the cut or raw edge of the duck. The number of pieces in the roll shall be indicated by stencil marking on the last 6 inches of the roll on the outer surface.

3.9 Fiber identification. Each roll of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Identification tickets. Each roll shall have a ticket attached with not finer than 5-ply cotton string doubled to not less than 6 inches long. The ticket shall be made of paperboard not less than 0.020 inch in thickness and the color shall be manila or light in intensity to permit easy reading of printed, stamped, or typed markings. The use of handwritten entries is prohibited. The ticket shall have clipped corners at the end where a reinforcing patch (with or without a metal eyelet) is firmly affixed for attaching the tying string. The ticket shall be legibly printed with the following information using water insoluble ink.

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STOCK NUMBER
NOMENCLATURE
SPECIFICATION NUMBER
YARDAGE
CONTRACT NUMBER AND DATE
MANUFACTURER'S NAME

3.11 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. The preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical properties in accordance with the methods specified in 4.5.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of 4.3.2.1 through 4.3.2.4.

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4.3.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section III of Fed Std No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.1.1 except as follows:

a. Only coarse yarns that exceed twice the normal yarn diameter shall be scored.

b. Mixed filling (shade bar) shall be scored only when resulting from wrong ply, wrong twist in the yarn, or off shade yarn.

c. Only knots and slubs which exceed limits shown in figure 1 of Fed Std No. 4 shall be scored.

No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 60 points. If one roll exceeds 60 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 60 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards}$$

4.3.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

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The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Uneven weaving

4.3.2.2 Examination for length

4.3.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.3.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.3.2.4 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133, to determine that packaging, packing and marking complies with the Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in Fed Std No. 191 wherever applicable, as listed in table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous

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yards, full width, of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size (number of sample units) shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II. Test methods

Tests	Fed Std No. 191 Method
Weight of cloth	5040, 5041
Yarns per inch	5050
Breaking strength	5104
Tearing strength	5134
Block 1/	5872

1/ Time of exposure shall be 24 hours at a temperature of 140 ± 1.8°F (60° ± 1°C). A six (6) pound weight shall be used.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.1.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.1.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.2 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use. This cloth is intended for use in the manufacture of seat cushion covers, armor vest covers, and equipage other than those mentioned above.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Color (see 3.5).

c. First article (see 3.2, 4.2 and 6.6).

d. Quantity and dimensions of cloth required (see 3.8.2).

e. Selection of applicable levels of packaging and packing as designated by the procuring activity (see 5.1).

f. Width tolerance (see 3.8.1).

g. Blocking test (see 3.7).

6.2.1 In the preparation of contracts and orders, it shall be noted that various options, choices, and alternatives as indicated in PPP-P-1133, may be exercised in the preparation of this cloth for delivery.

6.3 A 20/2 yarn has been found to be successful in producing this yarn. The yarn was produced with a 4.50 twist multiplier in the singles and a balanced twist in the ply.

6.4 The following finishing procedure has been found to be successful in producing this fabric:

a. Scour open width on continuous scouring range at 160°F in scouring solution of 1.0 g/l of TSP and detergent.

b. Dry at 300°F to set construction.

c. Autoclave heat set using procedure below:

5 minutes pre-vacuum
20 minutes 30 psi steam
5 minutes vacuum
20 minutes 30 psi steam
5 minutes vacuum
Exhaust to air.

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d. Pad on antistatic solution: 6.6 percent Aston 123*
2.7 percent Accelerator EN
pH - 6.5

Wet pick up 50 percent - 60 percent.

e. Dry at 300°F, cure at 350°F for 1.5 minutes.

* The requirements for the durable antistatic treatment can be met with Aston 123, Onyx Chemical Company, 190 Warren Street, Jersey City, New Jersey 07032. Other products considered for this use must have prior approval of the contracting officer.

6.5 A non-melting aromatic polyamide fiber is produced by E. I. du Pont de Nemours under the trademark NOMEX.

6.6 First article. The cloth covered by this specification requires first article inspection and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should consist of 5 yards of the dyed and finished cloth and shall be submitted or made ready for the contracting officer or his authorized representative. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, test and approval of the first article.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Project Number: 8305-F236

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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